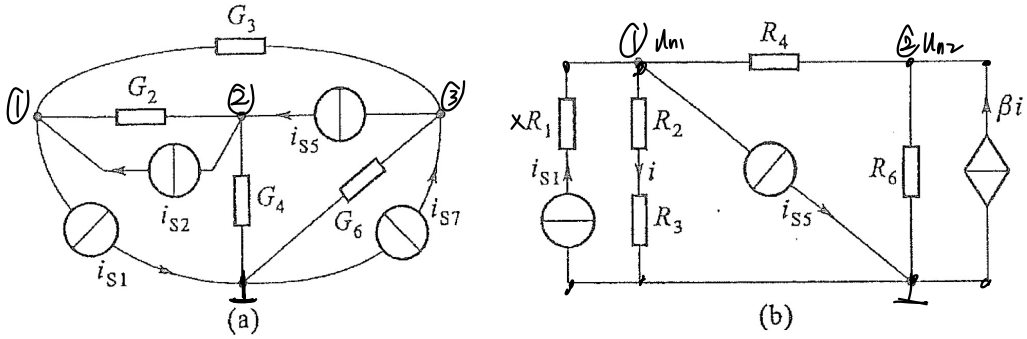
3-15 列出题 3-15 图(a)(b)所示电路的节点电压方程。



题 3-15 图

(a) $(G_2+G_3)U_{n1} - G_2U_{n2} - G_3U_{n3} = i_{S2} - i_{S1}$

$-G_2U_{n1} + (G_2+G_4)U_{n2} = i_{S5} - i_{S2}$

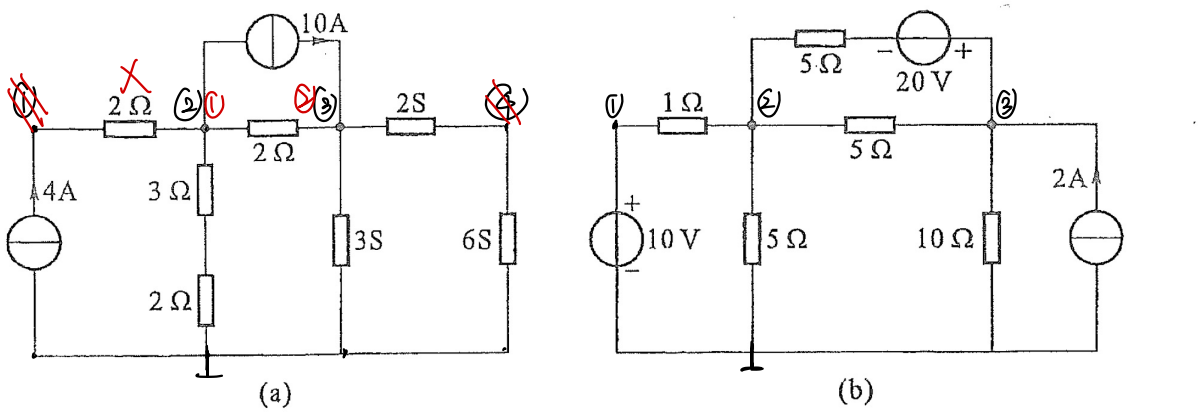
$-G_3U_{n1} + (G_3+G_6)U_{n3} = i_{S7} - i_{S5}$

(b) $\left(\frac{1}{R_2+R_3} + \frac{1}{R_4}\right)U_{n1} - \frac{1}{R_4}U_{n2} = i_{S1} - i_{S5}$

$-\frac{1}{R_4}U_{n1} + \left(\frac{1}{R_4} + \frac{1}{R_6}\right)U_{n2} = \beta i$

增补 $\frac{U_{n1}}{R_2+R_3} = i$

3-16 列出题 3-16 图(a)(b)所示电路的节点电压方程。



题 3-16 图

(a) $\frac{1}{2}U_{n1} - \frac{1}{2}U_{n2} = 4$

$-\frac{1}{2}U_{n1} + \left(\frac{1}{2} + \frac{1}{2} + \frac{1}{3+2}\right)U_{n2} - \frac{1}{2}U_{n3} = -10$

$-\frac{1}{2}U_{n2} + \left(\frac{1}{2} + \frac{1}{2S} + \frac{1}{3S}\right)U_{n3} - \frac{1}{2S}U_{n4} = 10$

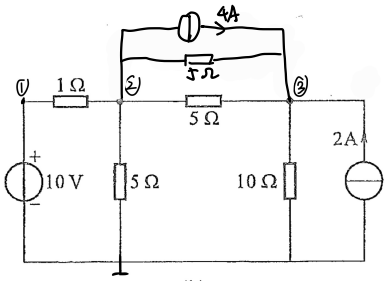
$-\frac{1}{2S}U_{n3} + \left(\frac{1}{2S} + \frac{1}{6S}\right)U_{n4} = 0$

$\left(\frac{1}{2} + \frac{1}{3+2}\right)U_{n1} - \frac{1}{2}U_{n2} = 4 - 10$

$-\frac{1}{2}U_{n1} + \left(\frac{1}{2} + 3 + \frac{1}{2+3}\right)U_{n2} = 10$

$\Rightarrow \begin{cases} 0.7U_{n1} - 0.5U_{n2} = -6 \\ -0.5U_{n1} + 10U_{n2} = 10 \end{cases}$

(b) 等效于

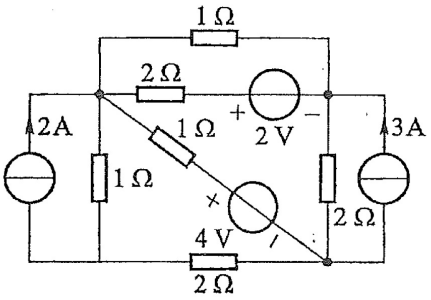


$U_{n1} = 10V$

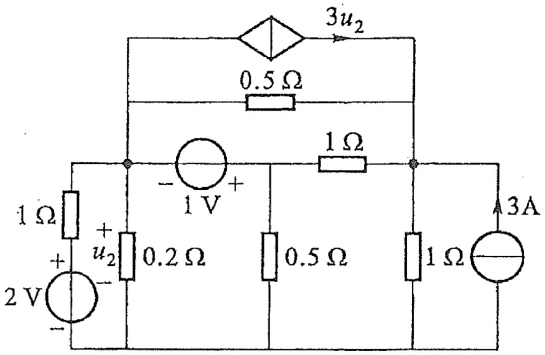
$-1 \times U_{n1} + \left(1 + \frac{1}{5} + \frac{2}{5}\right)U_{n2} - \frac{2}{5} \times U_{n3} = -4$

$-\frac{2}{5} \times U_{n2} + \left(\frac{2}{5} + \frac{1}{10}\right)U_{n3} = 4 + 2$

3-18 列出题 3-18 图(a)(b)所示电路的节点电压方程。



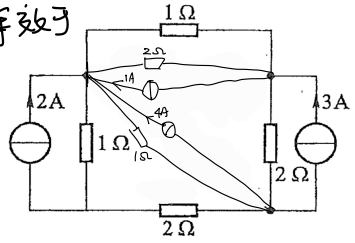
(a)

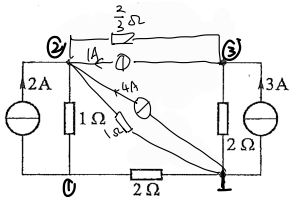


(b)

题 3-18 图

(a) 等效于



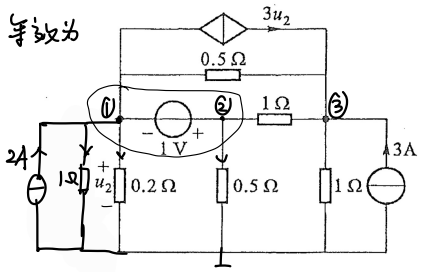


$(\frac{1}{2}+1)U_{n1}-U_{n2}=-2$

$-U_{n1}+(1+1+\frac{2}{3})U_{n2}-\frac{2}{3}U_{n3}=2+4+1$

$-\frac{2}{3}U_{n2}+(\frac{2}{3}+\frac{1}{2})U_{n3}=3-1$

(b) 等效为



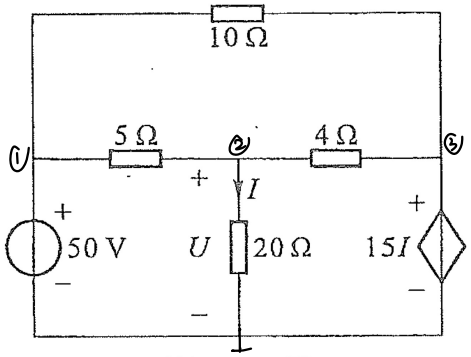
节点3：$-1\times U_{n2}-\frac{1}{0.5}U_{n1}+(\frac{1}{0.5}+1+1)U_{n3}=3+3U_2$

超级节点KCL：$2-\frac{U_{n1}}{1}-\frac{U_{n1}}{0.2}+\frac{U_{n3}-U_{n1}}{0.5}-3U_2-\frac{U_{n2}}{0.5}+\frac{U_{n3}-U_{n2}}{1}=0$

辅补：$U_2-U_{n1}=1V$

$U_{n1}=U_2$

3-21 用节点电压法求解题 3-21 图所示电路中电压 U。(请注意如何使待解的独立方程数为最少。)



题 3-21 图

$U_{n1}=50V$

$U_{n3}=15I$

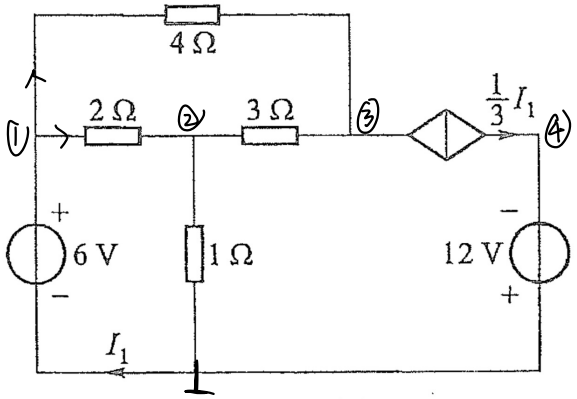
$-\frac{1}{5}U_{n1}+(\frac{1}{5}+\frac{1}{20}+\frac{1}{4})U_{n2}-\frac{1}{4}U_{n3}=0$

$I=\frac{U_{n2}}{20\Omega}$

整理⇒ $\begin{cases}-10+\frac{1}{2}U_{n2}-\frac{15}{4}I=0\\ I=\frac{U_{n2}}{20}\end{cases}$

$\Rightarrow U=U_{n2}=32V$

3-24 用节点电压法求解题 3-24 图所示电路后，求各元件的功率并检验功率是否平衡



题 3-24 图

$U_{n1}=6V$

$U_{n4}=-12V$

$-\frac{1}{2}U_{n1}+(\frac{1}{2}+1+\frac{1}{3})U_{n2}-\frac{1}{3}U_{n3}=0$

$-\frac{1}{4}U_{n1}-\frac{1}{3}U_{n2}+(\frac{1}{3}+\frac{1}{4})U_{n3}=-\frac{1}{3}I_1$

节点①KCL $I_1=\frac{U_{n1}-U_{n3}}{4}+\frac{U_{n1}-U_{n2}}{2}$

整理⇒ $-3+\frac{11}{6}U_{n2}-\frac{1}{3}U_{n3}=0$

$-\frac{3}{2}-\frac{1}{3}U_{n2}+\frac{7}{12}U_{n3}+\frac{1}{3}I_1=0$

$I_1=\frac{3}{2}-\frac{1}{4}U_{n3}+3-\frac{1}{2}U_{n2}$

$\Rightarrow U_{n2}=2V \quad U_{n3}=2V \quad I_1=3A$

6V电压源：$P_{发}=6V\times 3A=18W$（发出）

2Ω电阻：$P_{吸}=\frac{(6-2)^2}{2}=8W$（吸收）

1Ω电阻：$P_{吸}=\frac{2^2}{1}=4W$（吸收）

4Ω电阻：$P_{吸}=\frac{(6-2)^2}{4}=4W$（吸收）

3Ω电阻：$P_{吸}=0W$

电流源：$P_{吸}=[2-(-12)]\times\frac{1}{3}\times 3=14W$（吸收）

12V电压源：$P_{发}=\frac{1}{3}\times 3\times 12=12W$（发出）

验证：$P_{发总}=18+12=30W$

$P_{吸总}=8+4+4+0+14=30W$

平衡